Quantum Walk Search for Marked Vertices as a Model of Cognitive Search:

From Semantic Memory Retrieval to Multi-Alternative Decision Making

[Student Name 1] [Student ID 1]
[Student Name 2] [Student ID 2]

EE5105 Special Topics in Quantum Information Science
Instructor: Prof. Shih-Han Hung Fall 2026

Abstract

We use the Ambainis–Gilyén–Jeffery–Kokainis (AGJK) marked-vertex search algorithm as the technical core of a process-level model of cognitive search. Numerically, the quantum search time scales as $T_q \sim \mathrm{HT}^{0.44\text{–}0.52}$ on complete graphs, cycles, grids, and hypercubes, and on small-world and scale-free graphs with exponent 0.52. Every independent marked set is easier than every clustered one, while clustered sets can be easier or harder than a single target. The extended-hitting-time gap grows with size only in the AGJK adversarial torus construction (up to 8.8 at $L = 36$); one independent set on a barbell graph shows a gap of 2.9. On a toy semantic graph, continuous-time quantum walks produce larger and later target activation than classical diffusion, especially for cross-community targets. In a multi-alternative decision graph, coupling two options shifts the quantum choice probability toward the uncoupled option (from 1/3 to 0.50 on a symmetric three-option graph), whereas the classical choice stays uniform; this holds only under phase-free readouts, and we show that a first-threshold readout is dominated by the oscillation phase. A partial- measurement sweep shows that the pure walk has the highest peak but needs it to be timed; when a measurement can end the search, an intermediate rate ($p \approx 0.2$–0.3) finds clustered targets by a moderate deadline with higher probability than either the pure quantum or the classical walk (0.45 versus 0.33 at $T = 40$ on a grid). The project connects quantum-walk complexity to testable model-level predictions without claiming a biological quantum advantage.

1 Introduction

Quantum cognition models judgment, memory, and decision making with the mathematical structure of quantum probability. It does not claim that the brain is a quantum computer. Instead, it uses Hilbert spaces, superposition, and interference to capture contextuality, order effects, and evidence accumulation that classical probability handles awkwardly (Pothos and Busemeyer, 2022; Busemeyer and Bruza, 2024). Quantum walks are a natural process-level ingredient: a walker on a semantic or decision graph represents unfolding evidence, a marked vertex represents a target memory or an option, and the walk's statistics define response times and choice probabilities (Chen et al., 2022; Huang and Busemeyer, 2022; Pavone and Faro, 2025).

The missing link is algorithmic structure. The quantum-walk literature on search focuses on worst-case query and time complexity: given a graph and a marked set, how quickly can one detect or find a marked vertex? The quantum cognition literature focuses on explaining behavioral signatures. The two bodies of work meet mostly through the metaphor that a quantum walk is

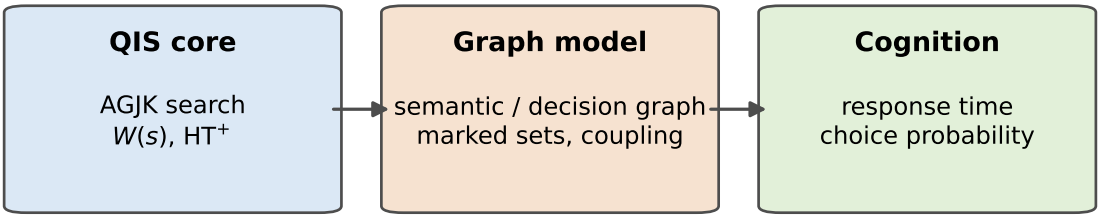


Figure 1: Three-layer architecture. The AGJK search core supplies the walk operator and the hitting-time structure; the graph layer maps concepts or options to vertices and marked sets; the cognitive layer turns walk statistics into response times and choice probabilities.

an evidence accumulator. The marked-vertex search problem itself—especially its behavior with multiple marked vertices—is rarely used as a formal model of cognitive search.

This project uses the Ambainis–Gilyén–Jeffery–Kokainis (AGJK) algorithm as the technical core (Ambainis et al., 2019). AGJK gives a quantum walk algorithm that finds a marked vertex on an arbitrary graph, for an arbitrary marked set, with a quadratic speedup over the classical hitting time up to logarithmic factors. It also isolates the extended hitting time HT^+, which can be much larger than HT when there are multiple marked vertices. That structure maps naturally onto cognitive search with several targets and onto multi-alternative choice, where similar options compete and context changes the choice distribution.

We ask three questions.

1. **Mapping.** How do HT, HT^+, the spectral gap, and the success probability map onto response time, choice probability, and deliberation speed on a semantic or decision graph?

2. **Structure.** How do the number and arrangement of marked vertices—single, independent, clustered—change the quantum versus classical search time and success probability? Can HT^+ formalize multi-alternative decision difficulty and context effects?

3. **Noise.** How does partial measurement or decoherence erode the quantum advantage, and is there an optimal "introspection" level that maximizes success probability or decision speed?

Figure 1 summarizes the three layers of the project: the QIS core, the graph model, and the cognitive quantities derived from the walk.

The project makes four contributions. First, we implement the Szegedy walk, the interpolated walk $W(s)$, and the AGJK extended hitting time, and we verify the $\widetilde{O}(\sqrt{\mathrm{HT}})$ scaling numerically on complete graphs, cycles, grids, and hypercubes, and on small-world and scale-free graphs. Second, we compare single, independent, and clustered marked sets. Every independent set is easier than every clustered set, while clustered sets can be easier or harder than a single target depending on their placement. The extended hitting time exceeds the classical one substantially only in special arrangements: it grows with size only for the adversarial periodic construction of AGJK, and a moderate gap of 2.9 appears for one independent set on a barbell graph. Third, we build a semantic graph and a multi-branch decision graph and turn the walk statistics into target activation, response time, and choice probability. Fourth, we add a partial-measurement model and show that single-deadline comparisons are dominated by the phase of the quantum oscillation. With deadline-free

readouts, the pure walk keeps the highest timed peak, while at moderate deadlines an intermediate measurement rate that can stop the search finds clustered targets faster than both the pure quantum and the classical walk.

The paper is organized as follows. Section 2 reviews classical and quantum walks and the quantum cognition literature. Section 3 collects the formal definitions and the AGJK results we use. Section 4 states the cognitive mapping and its assumptions. Section 5 describes the graph families, protocols, and metrics. Section 6 reports E1–E6. Section 7 discusses interpretation, limitations, and future work.

2 Background

2.1 Classical random walks and hitting times

Let $G = (V, E)$ be a connected graph with $n = |V|$ vertices and let P be a row-stochastic transition matrix. A probability row vector π is stationary if $\pi P = \pi$. For a reversible chain, detailed balance holds: $\pi_x P_{xy} = \pi_y P_{yx}$ for all x, y.

Given a non-empty marked set $M \subset V$, the hitting time is the expected number of steps needed to reach M from the stationary distribution,

$$\mathrm{HT}(P, M) = \mathbb{E}_{X_0 \sim \pi} \left[\min\{t \geq 0 : X_t \in M\} \right]. \tag{1}$$

For computations we use the equivalent conditioned form

$$\mathrm{HT}(P, M) = \frac{1}{1 - p_M} \sum_{x \notin M} \pi_x h_x, \qquad h_x = 1 + \sum_y P_{xy} h_y, \quad x \notin M, \tag{2}$$

with $p_M = \sum_{x \in M} \pi_x$. The linear system in (2) is small for the graphs in this paper and can be solved directly. Standard examples are $\mathrm{HT} = \Theta(N)$ on the complete graph, $\mathrm{HT} = \Theta(N^2)$ on the cycle, and $\mathrm{HT} = \Theta(N \log N)$ on the two-dimensional grid; the hypercube has $\mathrm{HT} = \Theta(N)$, and our data ($N = 8$–64) give $\mathrm{HT}/N \approx 1.15$–$1.34$.

2.2 Szegedy walks and the detection–finding gap

A Szegedy walk lifts a Markov chain to a unitary operator on two vertex registers. With the convention used by AGJK,

$$V(P)|\bar{0}\rangle|x\rangle = \sum_y \sqrt{P_{xy}}|y, x\rangle, \qquad W(P) = V^\dagger(P)\,\mathrm{Shift}\,V(P)\,\mathrm{Ref}, \tag{3}$$

where $\mathrm{Shift}|x, y\rangle = |y, x\rangle$ and $\mathrm{Ref} = (2|\bar{0}\rangle\langle\bar{0}| - I) \otimes I$. The discriminant matrix is

$$D_{xy} = \sqrt{P_{xy} P_{yx}} = \langle\bar{0}|\langle x|W(P)|\bar{0}\rangle|y\rangle. \tag{4}$$

For a reversible chain, $D = \mathrm{diag}(\pi)^{1/2} P\, \mathrm{diag}(\pi)^{-1/2}$, so D and P share their spectrum and D is symmetric.

Szegedy's search algorithm detects a marked vertex in $O(\sqrt{\mathrm{HT}})$ steps. Detection is not the same as finding: measuring the final state need not return a marked vertex. The gap is especially visible when there are multiple marked vertices. This is the starting point of AGJK.

Paper	Model	Cognitive phenomenon	Role here
AGJK 2019	quantum walk search	marked-vertex search, HT^+	QIS core
Chen et al. 2022	DTQW	evidence accumulation, decoherence	measurement model
Huang–Busemeyer 2022	quantum walk	multi-alternative choice	decision mapping
Pavone–Faro 2025	CTQW on semantic graphs	insight, interference	semantic experiment
Pothos–Busemeyer 2022	quantum probability	context, order effects	motivation
Epping et al. 2023	open-system model	choice and response time	noise interpretation

Table 1: Literature map. The technical core is AGJK; the other papers supply the cognitive interpretation and the measurement/noise models.

2.3 The AGJK algorithm and extended hitting time

AGJK introduce the absorbing walk P' and the interpolated walk $P(s) = (1-s)P + sP'$, which stays at a marked vertex with probability s instead of moving. The corresponding quantum walk $W(s)$ uses an ancilla to record whether the current vertex is marked. AGJK define the extended hitting time HT^+ through the eigenvalues and eigenvectors of the discriminant matrix $D(s)$; if there is a single marked vertex, $\mathrm{HT}^+ = \mathrm{HT}$, but with multiple marked vertices HT^+ can be much larger.

Two algorithms are given. The simpler one runs $W(s)$ for $O(\sqrt{\mathrm{HT}})$ steps and is supported by a numerical conjecture (Conjecture 3 in the paper): for some s and $t = O(\sqrt{\mathrm{HT}})$, measuring the vertex register returns a marked vertex with constant probability. The proved algorithm combines the interpolated walk with quantum fast-forwarding and achieves $\widetilde{O}(\sqrt{\mathrm{HT}})$ time for any graph and any marked set (Ambainis et al., 2019). We implement and test the first algorithm numerically; we cite the second as the theoretical guarantee and do not re-prove it.

2.4 Quantum cognition and quantum walks

Quantum cognition replaces classical probability with quantum probability in models of judgment and decision. Pothos and Busemeyer review the program and its explanations of order effects, conjunction and disjunction effects, and contextuality (Pothos and Busemeyer, 2022; Busemeyer and Bruza, 2024). The models are usually process-level: they do not assume that the brain implements a quantum circuit.

Quantum walks enter this literature in several ways. Chen, Ferro, and Sornette use discrete-time quantum walks as evidence accumulators and discuss how partial measurement produces decoherence (Chen et al., 2022). Huang and Busemeyer propose a quantum walk framework for multi-alternative decision making (Huang and Busemeyer, 2022). Pavone and Faro use continuous-time quantum walks on semantic graphs to model the emergence of insight through interference (Pavone and Faro, 2025). Epping et al. develop an open-system model of choice and response time (Epping et al., 2023), and Widdows et al. give quantum circuit components for cognitive decision-making (Widdows et al., 2023).

2.5 The gap

The two literatures meet at the level of metaphor but not at the level of algorithmic structure. Quantum-walk search papers analyze worst-case complexity and spectral quantities but do not ask what those quantities mean for a cognitive process. Quantum-cognition papers use walks as

evidence accumulators but rarely exploit the marked-set structure, the extended hitting time, or the difference between detection and finding. This paper treats the marked-vertex search problem as a process model and uses its structural quantities—HT, HT^+, the spectral gap, and the success curve—as the bridge between the two literatures.

3 Theory

3.1 Reversible chains and the discriminant matrix

Let P be an ergodic reversible Markov chain with stationary distribution π. The discriminant matrix is

$$D_{xy} = \sqrt{P_{xy}P_{yx}}, \qquad D = \mathrm{diag}(\pi)^{1/2} P \,\mathrm{diag}(\pi)^{-1/2}. \tag{5}$$

For reversible P, D is symmetric and has the same spectrum as P; the vector $\sqrt{\pi}$ is a left eigenvector with eigenvalue 1. The Szegedy walk (3) satisfies

$$\langle\bar{0}|\langle x|W(P)|\bar{0}\rangle|y\rangle = D_{xy}. \tag{6}$$

The following standard spectral fact is used repeatedly. If $D = \sum_k \lambda_k |v_k\rangle\langle v_k|$, then the nontrivial eigenvalues of $W(P)$ are $e^{\pm i\theta_k}$ with $\lambda_k = \cos\theta_k$, and the corresponding eigenvectors lie in the span of $\{|\bar{0}\rangle|x\rangle\}_{x\in V}$ and its image under $V(P)$. This is the Szegedy spectral lemma; we use it as a statement and do not reprove it.

3.2 Classical and extended hitting times

For a marked set M with stationary mass $p_M = \sum_{x\in M}\pi_x$, the classical hitting time can be written in spectral form as

$$\mathrm{HT}(P,M) = \frac{1}{1-p_M}\sum_{k=1}^{n-|M|}\frac{|\langle v'_k|\sqrt{\pi_U}\rangle|^2}{1-\lambda'_k}, \tag{7}$$

where λ'_k are the non-unit eigenvalues of $D(1)$ and $\sqrt{\pi_U}$ is the subnormalized vector with entries $\sqrt{\pi_x}$ for $x \notin M$.

AGJK replace D by the discriminant matrix $D(s)$ of the interpolated walk and define

$$\mathrm{HT}(s) = \frac{1}{1-p_M}\sum_{k=1}^{n-1}\frac{|\langle v_k(s)|\sqrt{\pi_U}\rangle|^2}{1-\lambda_k(s)}, \qquad \mathrm{HT}^+ = \lim_{s\to 1}\mathrm{HT}(s), \tag{8}$$

where $D(s) = \sum_k \lambda_k(s)|v_k(s)\rangle\langle v_k(s)|$ and $\lambda_n(s) = 1$. For a single marked vertex, $\mathrm{HT}^+ = \mathrm{HT}$. For multiple marked vertices, (8) can be much larger than (7).

The canonical example is an $N \times N$ torus with marked set

$$M = M_1 \cup M_2, \qquad M_1 = \{(j_1 d_1, j_2 d_1) : 0 \le j_1, j_2 < k_1\}, \quad M_2 = \{(j_1 d, j_2 d) : 0 \le j_1, j_2 < N/d\}, \tag{9}$$

with $d_1 \mid d \mid N$ and the scale separation of AGJK. The set M_1 is a dense cluster and M_2 is a sparse periodic grid. For suitable parameters, $\mathrm{HT} = \widetilde{O}(d^2)$ while $\mathrm{HT}^+ = \Omega(N^2)$; this is AGJK's asymptotic statement. Our numerics ($L = 18$–36) can show growth of $\mathrm{HT}^+/\mathrm{HT}$ but cannot establish the asymptotic rate. The large gap comes from the periodic structure, not from the mere fact that $|M| > 1$.

3.3 Interpolated walk and the numerical AGJK protocol

Let $U = V \setminus M$. The absorbing and interpolated walks are

$$P' = \begin{pmatrix} P_{UU} & P_{UM} \\ 0 & I \end{pmatrix}, \qquad P(s) = (1-s)P + sP', \quad s \in [0,1). \tag{10}$$

The quantum version attaches a qubit that records whether the current vertex is marked. For $x \in U$, $V(P,s)$ acts as $V(P)$; for $x \in M$ it maps

$$|0\rangle|\bar{0}\rangle|x\rangle \mapsto \sqrt{1-s}|0\rangle V(P)|\bar{0}\rangle|x\rangle + \sqrt{s}|1\rangle|\bar{0}\rangle|x\rangle. \tag{11}$$

The interpolated walk operator is

$$W(s) = V^\dagger(P,s)\,\mathrm{Shift}'\,V(P,s)\,\mathrm{Ref}', \tag{12}$$

with the controlled shift and reflection of AGJK. Its top-left block is $D(s)$.

Algorithm 1 of AGJK prepares $|\bar{0}\rangle|\sqrt{\pi}\rangle$, applies $W(s)$ for t steps, and measures the marking ancilla. If the ancilla is 1, a position measurement returns a candidate marked vertex. The key numerical object is the lower bound

$$q_t(s) = \left\|\Pi_M T_t(D(s))|\sqrt{\pi}\rangle\right\|^2 \le \left\|(I \otimes \Pi_M)W^t(s)|\bar{0}\rangle|\sqrt{\pi}\rangle\right\|^2, \tag{13}$$

where T_t is the Chebyshev polynomial of the first kind. The identity behind (13) is

$$\Pi_0 W^t(s)\Pi_0 = |\bar{0}\rangle\langle\bar{0}| \otimes T_t(D(s)), \qquad \Pi_0 = |\bar{0}\rangle\langle\bar{0}| \otimes I. \tag{14}$$

In our numerical protocol we scan s over a fixed grid and choose the value that maximizes $\max_t q_t(s)$ over $t \le 3\sqrt{\mathrm{HT}} + 5$. We then run the exact operator $W(s)$ and record the actual success curve $p_M(t)$. This tests the numerical content of Conjecture 3 of AGJK without assuming it.

For completeness, the proved AGJK algorithm combines the interpolated walk with quantum fast-forwarding and runs in time

$$O\Big(S\sqrt{\log \mathrm{HT}} + \sqrt{\mathrm{HT}}(U+C)\sqrt{\log \mathrm{HT} \log\log \mathrm{HT}}\Big), \tag{15}$$

where S, U, and C are the costs of setup, update, and checking. We cite (15) as the theoretical guarantee; our experiments use the simpler Algorithm 1.

3.4 Partial measurement

The decoherence model is a position measurement with probability p before each unitary step. Let $\Pi_x = I \otimes |x\rangle\langle x|$ be the projector onto vertex x and let $R_x = |\bar{0}\rangle|x\rangle$ be the reset state. The one-step channel is

$$\rho_{t+1} = W(0)\left[(1-p)\rho_t + p\sum_x R_x\langle R_x|\rho_t|R_x\rangle\langle R_x|\right]W(0)^\dagger. \tag{16}$$

For $p = 0$ this is the pure quantum walk. For $p = 1$ the state is reset to a position basis state at every step; the position distribution then follows the classical Markov chain

$$Q_{xy} = \left\|\Pi_y W(0)|\bar{0}\rangle|x\rangle\right\|^2. \tag{17}$$

For $s = 0$, $Q = P$ and the measured walk is exactly the classical random walk.

Walk object	Cognitive interpretation
Vertex $x \in V$	concept, evidence state, or option
Edge $(x, y) \in E$	semantic association or evidence transition
Marked set M	target memory or decision boundary
Initial state $\|\psi_0\rangle$	cue, question, or initial attention
Unitary $W(s)$ or e^{-iLt}	deliberation / evidence accumulation
Measurement	committing to an answer or decision
$p_M(t)$	target activation or success probability
$T_q = \arg\max_t p_M(t)$	response time at peak evidence
First threshold crossing	response time under a deadline
P_i for option sets M_i	choice probability
HT, HT$^+$	expected search difficulty, context difficulty
Spectral gap λ_1	deliberation / relaxation speed
Interpolation s	commitment or absorption strength
Measurement rate p	introspection or noise level

Table 2: Mapping between walk quantities and cognitive quantities. The mapping is process-level and mathematical; it is not a claim about the physical implementation of cognition.

Because (16) is linear in ρ, the average success probability can be computed without Monte Carlo. We precompute the pure states $|\phi_{x,a}\rangle = W(0)^a|\bar{0}\rangle|x\rangle$ and their position marginals and marked masses. A trajectory is summarized by the probability q_0 that no measurement has occurred and by the distribution $q_{x,a}$ over the last reset vertex x and age a. The update is a small linear recursion; the success probability is the corresponding mixture of the precomputed marked masses. The details are in appendix C.

3.5 Continuous-time walks on semantic graphs

For the semantic experiment we use the symmetric normalized Laplacian

$$L_{\text{sym}} = I - D^{-1/2} A D^{-1/2} \tag{18}$$

and the random-walk generator $L_{\text{rw}} = I - D^{-1}A$. The CTQW and CRW are

$$|\psi(t)\rangle = e^{-iL_{\text{sym}}t}|c\rangle, \qquad \rho(t) = \rho(0)e^{-L_{\text{rw}}t}, \tag{19}$$

where c is the cue vertex. The eigenvalues of L_{sym} and L_{rw} coincide, so comparing at $\tau = \lambda_1 t$ aligns the two relaxation scales. The target activation is $p_T(t) = |\langle T|\psi(t)\rangle|^2$ for CTQW and $p_T(t) = \rho_T(t)$ for CRW.

4 Cognitive Model

4.1 From walk objects to cognitive quantities

The model treats a graph as a semantic or decision network and the walk as a deliberation process. Table 2 lists the correspondence. The mapping is deliberately minimal: every cognitive quantity is a statistic of the walk, so the model can be simulated and compared with a classical baseline without additional free parameters.

4.2 Modeling assumptions

We make five assumptions explicit.

1. **Process level.** The model describes the statistics of a search or decision process. It does not claim that the brain implements a quantum walk or a unitary circuit.

2. **Interference as multiple paths.** Quantum interference stands for the interaction of multiple evidence paths or semantic associations. The phase structure is a modeling device, not a physiological statement.

3. **Classical baseline.** The baseline is a Markov random walk on the same graph with the same initial distribution and the same marked set. This isolates the effect of interference from the effect of graph structure.

4. **Defined, not fitted, statistics.** Response time, choice probability, target activation, entropy, and radius are defined from the walk statistics. We do not fit behavioral data in this paper.

5. **Graph as prior.** The graph encodes the semantic or decision prior. Changing edges or weights changes the predicted context effects.

4.3 Response time and choice probability

For a single target, the activation curve is $p_M(t)$. We use two response-time definitions. The *peak time* is $T_q = \arg\max_{t \geq 1} p_M(t)$, the time at which the evidence for the target is strongest. The *threshold time* is the first t with $p_M(t) \geq \theta$, where θ is a decision threshold. The threshold definition is closer to a deadline experiment; the peak definition is closer to an unconstrained search. For decoherence we report the peak success probability, the success averaged over a uniform deadline, and the probability that a measurement has already detected the target by the deadline.

For multiple option sets $M_1, \ldots, M_k$, the option probabilities at time t are

$$P_i(t) = \left\| (I \otimes \Pi_{M_i}) W^t(s) |\psi_0\rangle \right\|^2, \qquad P_i = \frac{P_i(T)}{\sum_j P_j(T)}, \tag{20}$$

where T is the decision time. Because $P_i(t)$ oscillates, T must come from a phase-free stopping rule (a random stopping time or a measurement that detects an option, section 5.5); a first-threshold rule makes the choice depend on the oscillation phase. The choice entropy $-\sum_i P_i \log P_i$ measures how concentrated the choice distribution is. A uniform distribution gives maximal entropy; a strong context effect gives low entropy and a large $\max_i P_i - 1/k$ bias.

4.4 Similarity and context effects

The decision experiments manipulate similarity by coupling two branches at matching depths with weight c. A large c means that the two options are strongly associated: a walker can move between them without returning to the hub. In a classical random walk this mainly redistributes probability among the coupled options; on a symmetric graph the classical choice is unchanged. In a quantum walk the coupling changes the interference pattern, so the uncoupled option gains share. This is a model-level analogue of the similarity effect: the choice probability of an option depends on the presence and similarity of the other options, not only on its own value.

The semantic experiments manipulate distance in the same spirit. A cue and a target in the same community are close in the graph; a cross-community target requires a longer path. The

CTQW and CRW are expected to differ most when the target is far from the cue, because the quantum walk maintains coherent amplitude across several paths while the classical walk relaxes locally.

4.5 Testable predictions

The model yields four predictions that can be tested numerically and, in principle, behaviorally.

1. For well-connected graphs and a single target, quantum search time scales as $\sqrt{\mathrm{HT}}$ and beats the classical baseline.

2. Every independent target set is easier than every clustered one; a large HT^{+} penalty requires a structured marked set rather than clustering alone.

3. Semantic distance changes activation timing; the quantum walk has a higher peak for far targets, while the classical walk has a higher early AUC for near targets.

4. Coherence and measurement trade off: the pure walk has the highest timed peak, while a partially measured walk that can stop on detection finds extended (clustered) targets faster than both the pure and the classical walk at moderate deadlines.

These predictions are model-level. Confirming them numerically does not show that human cognition is quantum; it shows that the quantum-walk model makes distinct, falsifiable predictions relative to its classical counterpart.

5 Methods

5.1 Graph families and marked sets

We use four synthetic graph families for the scaling experiments: complete graphs K_N, cycles C_N, two-dimensional grids $L \times L$, and hypercubes Q_d with $N = 2^d$. The multiple-target experiments add barbell graphs (two cliques joined by a path) and $N \times N$ torus graphs with the marked-set pattern used in the AGJK counterexample. The semantic experiment uses a hand-built 19-vertex concept graph with animal, vehicle, and instrument communities, connected by the bridge concepts `bird` and `airplane`. Decision experiments use a hub with one path per option; matching depths of two branches can be coupled with a tunable weight.

For a graph $G = (V, E)$ we form the row-stochastic random-walk matrix

$$P_{xy} = \frac{A_{xy}}{\sum_z A_{xz}}, \tag{21}$$

where A is the adjacency matrix. All Szegedy and AGJK walks (E1, E2, E4, E5) use the lazy walk $P \mapsto (1-\eta)P + \eta I$ with $\eta = 0.1$; the AGJK adversarial torus uses $\eta = 0.2$, following the original construction. Laziness removes the eigenvalue -1 of bipartite graphs and multiplies every hitting time by exactly $1/(1-\eta)$, so it does not change any scaling law. For the lazy walk we report the phase gap $1 - \lambda_2$ of the discriminant matrix, because on bipartite graphs the absolute gap $1 - \max_{i \geq 2}|\lambda_i|$ is capped at 2η by the image $2\eta - 1$ of the eigenvalue -1. (When $\lambda_2 < 0$, as on small complete graphs, the phase gap exceeds 1.) A marked set $M \subset V$ is either a single target, an independent set of k vertices chosen by farthest-point sampling, a connected cluster grown by BFS, or a set of option terminals. Table 3 lists the experiment matrix.

ID	Graph / system	Marked set	Noise	Main metric
E1	K_N, C_N, grid, Q_d	single	none	scaling of T_q and HT
E2	grid, barbell	single / independent / clustered	none	$p_M(t)$, HT^+
E2b	adversarial torus	AGJK pattern	none	$\mathrm{HT}^+/\mathrm{HT}$
E3	toy semantic graph	single target	none	activation, AUC
E4	multi-branch graph	option terminals	none	choice prob., RT
E5	grid, toy semantic	single, clustered	partial measurement	timed peak, detection by T
E6	Watts–Strogatz, Barabási–Albert	hub, peripheral	none	T_q vs HT, speedup

Table 3: Experiment matrix. Each row corresponds to a processed result file in `code/results/processed/`.

5.2 Classical hitting time

For a marked set M, let h_x be the expected time to reach M from $x \notin M$, with $h_x = 0$ for $x \in M$. The hitting times solve the linear system

$$h_x = 1 + \sum_y P_{xy} h_y, \qquad x \notin M. \tag{22}$$

The classical hitting time is the stationary average conditioned on starting outside M,

$$\mathrm{HT}(P, M) = \frac{\sum_{x \notin M} \pi_x h_x}{1 - p_M}, \qquad p_M = \sum_{x \in M} \pi_x. \tag{23}$$

We solve (22) directly for all grids and barbell graphs; the synthetic families are small enough that dense linear algebra is inexpensive.

5.3 Szegedy and AGJK search protocol

The quantum walk uses the register convention of AGJK,

$$V(P)|\bar{0}\rangle|x\rangle = \sum_y \sqrt{P_{xy}}|y, x\rangle, \qquad W(P) = V^\dagger(P)\,\mathrm{Shift}\,V(P)\,\mathrm{Ref}, \tag{24}$$

with $\mathrm{Shift}|x, y\rangle = |y, x\rangle$ and $\mathrm{Ref} = (2|\bar{0}\rangle\langle\bar{0}| - I) \otimes I$. The discriminant matrix is $D_{xy} = \sqrt{P_{xy}P_{yx}}$.

For multiple or clustered targets we use the interpolated walk

$$P(s) = (1 - s)P + sP', \qquad P' = \begin{pmatrix} P_{UU} & P_{UM} \\ 0 & I \end{pmatrix}, \tag{25}$$

and the corresponding walk operator $W(s)$ with an ancilla that records whether the current vertex is marked. The initial state is $|\bar{0}\rangle|\sqrt{\pi}\rangle$. For each graph and marked set we scan s over a fixed grid and select the value that maximizes the AGJK lower bound

$$q_t(s) = \left\|\Pi_M T_t(D(s))|\sqrt{\pi}\rangle\right\|^2, \qquad t \le 3\sqrt{\mathrm{HT}} + 5, \tag{26}$$

where T_t is the Chebyshev polynomial of the first kind. We then run the actual walk $W(s)$ and record

$$p_M(t) = \left\|(I \otimes \Pi_M)W^t(s)|\bar{0}\rangle|\sqrt{\pi}\rangle\right\|^2. \tag{27}$$

The quantum search time is $T_q = \arg\max_{t\geq 1} p_M(t)$ and the peak success probability is $p_{\max} = \max_t p_M(t)$. The extended hitting time HT^+ is computed from the spectral definition

$$\mathrm{HT}(s) = \frac{1}{1-p_M} \sum_{k=1}^{n-1} \frac{|\langle v_k(s)|\sqrt{\pi_U}\rangle|^2}{1-\lambda_k(s)}, \qquad \mathrm{HT}^+ = \lim_{s\to 1} \mathrm{HT}(s), \tag{28}$$

evaluated at $s = 1 - 10^{-6}$ as a numerical proxy. Equation (28) is used only for diagnostics; the search curves use the actual operator $W(s)$.

5.4 Semantic activation protocol

For the semantic experiment we use a continuous-time quantum walk (CTQW) and a continuous-time random walk (CRW) from the same cue vertex. Let

$$L_{\mathrm{sym}} = I - D^{-1/2} A D^{-1/2} \tag{29}$$

be the symmetric normalized Laplacian and $L_{\mathrm{rw}} = I - D^{-1}A$ the random-walk generator. The two models are

$$|\psi(t)\rangle = e^{-iL_{\mathrm{sym}}t}|c\rangle, \qquad \rho(t) = \rho(0)e^{-L_{\mathrm{rw}}t}, \tag{30}$$

with $\rho(0) = |c\rangle\langle c|$ at the cue c. Time is reported in units of the spectral gap, $\tau = \lambda_1 t$, so that both walks are compared at the same relaxation scale. The target activation is $p_T(t) = |\langle T|\psi(t)\rangle|^2$ for CTQW and $p_T(t) = \rho_T(t)$ for CRW. We record the peak activation, time of peak, area under the activation curve, entropy at the peak, half-peak coverage, and the expected graph distance from the cue at the peak.

5.5 Decision protocol

The decision graph has a central hub and one path per option. Each option i corresponds to a terminal set $M_i = \{\mathrm{terminal}_i\}$. To control similarity, matching depths of branches i and j are joined by an edge of weight c; $c = 0$ gives independent branches and large c makes the two options highly similar. The quantum model runs the interpolated walk $W(s)$ with $s = 0.9$ and initial state $|\bar{0}\rangle|\mathrm{hub}\rangle$. The classical model runs the same graph with all option terminals absorbing. For each model we compute the option probabilities $p_i(t)$ and the total decision probability $P_M(t) = \sum_i p_i(t)$, with $T_{\max} = 400$. We compare four readouts. (i) *First crossing* (our original readout): choice probabilities are the normalized $p_i(t)$ at the first t with $P_M(t) \geq 0.5$, or at $\arg\max_t P_M(t)$ if the threshold is never reached. (ii) *Random stopping*: $P_i \propto T_{\max}^{-1} \sum_{t=1}^{T_{\max}} p_i(t)$. (iii) *Detection*: the measurement model of section 5.6 with rate $p = 0.2$, where a measurement landing on terminal i ends the decision with choice i; P_i is the probability of choice i by $T_{\max}$ (essentially one) and the decision time is the mean first-detection time. (iv) *Classical*: absorption probabilities and mean absorption time of the classical walk. To test robustness we repeat every configuration for $\eta \in \{0, 0.05, 0.1, 0.2\}$ and for ten independent draws in which every edge weight is multiplied by a factor uniform in $[0.8, 1.2]$, and report the spread (max minus min) of the probability of the uncoupled option(s).

5.6 Partial-measurement protocol

The decoherence experiment uses the Szegedy walk $W(0)$ of the lazy chain P_η ($\eta = 0.1$), i.e. interpolation $s = 0$, from a cue vertex. Before each step, with probability p we measure the vertex register in the position basis and reset the state to $|\bar{0}\rangle|x\rangle$; with probability $1 - p$ the state evolves

unitarily. Thus $p = 0$ is the pure quantum walk and $p = 1$ is exactly the classical random walk on the same graph. We compute the exact average over measurement trajectories by precomputing the position marginals and marked probabilities of the basis-state ensembles $W(0)^a|\bar{0}\rangle|x\rangle$, then propagating the age distribution of the most recent measurement. This avoids Monte Carlo noise and makes the sweep deterministic. The success probability is the probability that the walker is at a marked vertex after the step. We use $T_{\max} = 120$ and $p \in \{0, 0.05, 0.1, 0.2, 0.3, 0.5, 0.75, 1\}$.

Because $p_M(t)$ oscillates at small p, we do not compare rates at a single deadline. In the *readout model* we report the peak, the time-averaged success $\bar{p}(H) = H^{-1} \sum_{t=1}^{H} p_M(t)$ (the success under a uniform deadline on $\{1, \ldots, H\}$, the same device as randomized stopping in quantum search with unknown optimal time), and the coefficient of variation of p_M. In the *detection model*, a measurement that lands on M ends the search; the recursion is the same except that this mass is moved to a "found" counter instead of being reset (appendix C). With a final readout at the deadline T, $F_p(T)$ is the probability that the target has been found by T; F_0 equals the pure readout curve and $F_1(T) = \Pr[\tau_M \leq T]$ for the classical walk. We report $F_p(T)$ for $T \in \{20, 40, 80, 120\}$ and the best rate $p^*(T) = \arg\max_p F_p(T)$.

5.7 Metrics and reproducibility

The main metrics are the classical hitting time HT, the extended hitting time HT^+, the quantum search time T_q, the peak success probability $p_{\max}$, the speedup HT/T_q, the target AUC, the activation entropy and radius, choice probabilities, decision time, and the success probability at a fixed deadline. Scaling exponents are fitted by ordinary least squares in log–log coordinates.

The implementation is in Python 3.12 with NumPy, SciPy, NetworkX, Matplotlib, PyYAML, and pytest. The environment is created with the Miniforge file `code/environment.yml` (environment `qwalk-cognition`). All random choices use explicit seeds; the configuration files are `code/configs/base.yaml` and `code/configs/experiments.yaml`. The test suite contains 71 tests covering unitarity, norm preservation, the classical limit, hitting-time identities, decision normalization, and the exact measurement ensemble. Every figure in this paper is regenerated by the commands listed in `code/README.md`.

6 Results

6.1 Classical versus quantum scaling (E1)

Figure 2 shows classical hitting times and quantum search times for the four synthetic families. The quantum time is the time of peak success probability, and the interpolation parameter is selected by (26). The fitted exponents are collected in table 4. Across all 16 instances the peak success probability is between 0.84 and 1.00, so the reported T_q values are not driven by low-probability fluctuations.

All four exponents of T_q against HT have 95% confidence intervals that contain 1/2: complete 0.50 $[0.41, 0.59]$, cycle 0.52 $[0.51, 0.54]$, grid 0.48 $[0.39, 0.57]$, and hypercube 0.44 $[0.32, 0.55]$. The intervals come from a t-distribution with $N-2$ degrees of freedom and are wide because each family has only four sizes.

The cycle is the clearest illustration: its classical hitting time grows almost quadratically ($\alpha = 1.95$), while the quantum search time grows almost linearly ($\beta = 1.02$), giving a speedup that reaches $11.7\times$ at $N = 64$. The complete graph ($\mathrm{HT} = \Theta(N)$ exactly) and the hypercube both have nearly linear hitting times over this size range ($\alpha \approx 1.06$–1.07), and the quantum exponents are again close to 1/2. The grid shows a slightly larger exponent because the finite-size range does not

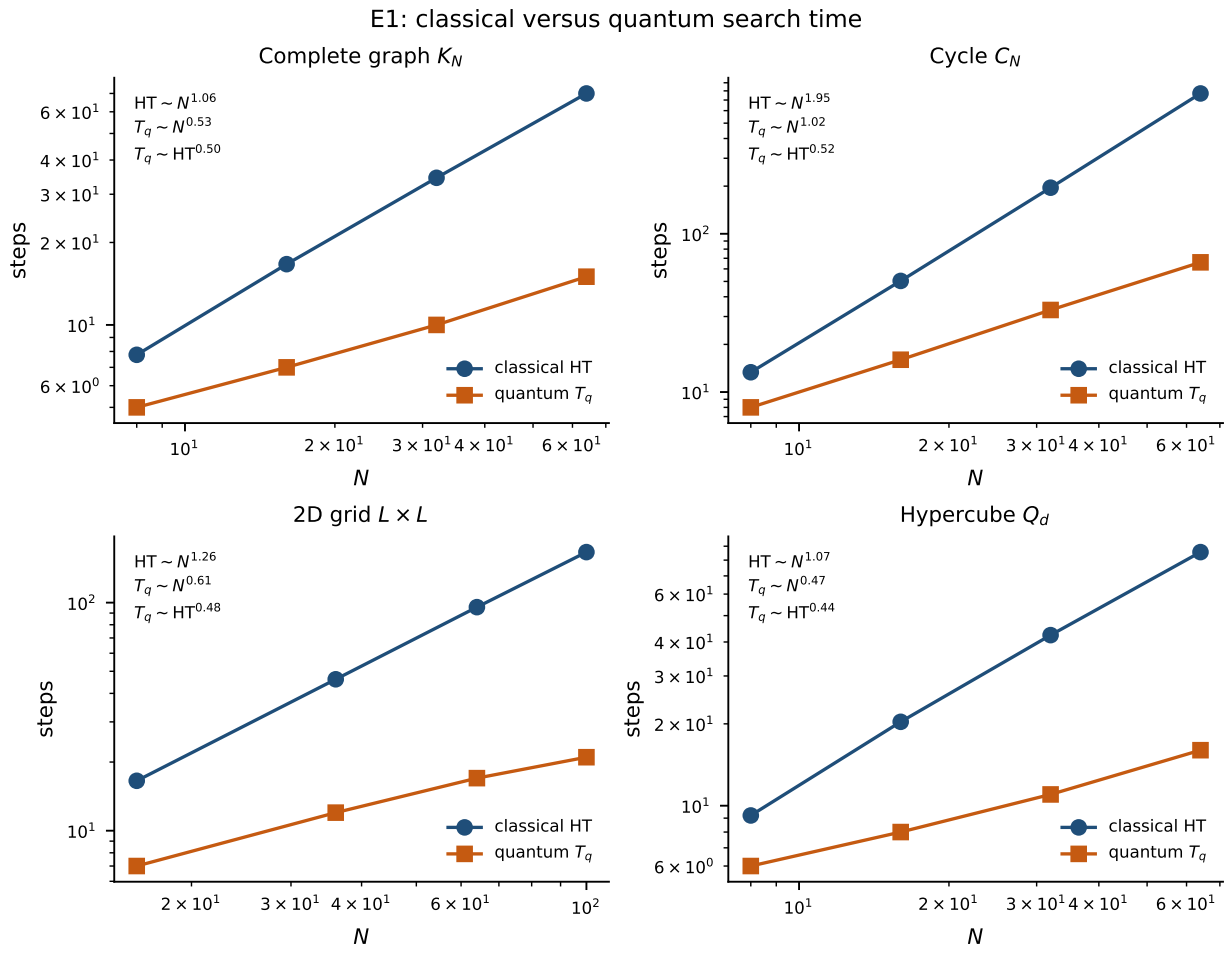


Figure 2: Classical hitting time and quantum search time versus graph size for complete graphs, cycles, 2D grids, and hypercubes. Both axes are logarithmic; lines are power-law fits. The quantum time tracks a square-root law in the classical hitting time rather than in the raw vertex count.

yet fully separate N from $\log N$; the fit against HT gives $\gamma = 0.48$, consistent with the predicted square-root law. The hypercube has the lowest exponent ($\gamma = 0.44$); with only four sizes and integer $T_q \in \{6, 8, 11, 16\}$ this fit is the least precise of the four.

Family	HT $\sim N^{\alpha}$	$T_q \sim N^{\beta}$	$T_q \sim \mathrm{HT}^{\gamma}$	max HT/T_q
Complete K_N	1.06	0.53	0.50	4.7
Cycle C_N	1.95	1.02	0.52	11.7
2D grid $L \times L$	1.26	0.61	0.48	7.9
Hypercube Q_d	1.07	0.47	0.44	5.4

Table 4: Fitted scaling exponents from E1. The exponent γ is the relevant comparison for the AGJK guarantee: it is close to 1/2 in all four families.

6.2 Single, independent, and clustered targets (E2)

Figure 3 compares success-probability curves for single, independent, and clustered marked sets. Each configuration uses every distinct set produced by farthest-point or BFS sampling with seeds $0, \ldots, 19$; this gives 14–18 distinct sets per configuration (the single target has one). Every independent set is easier than every clustered set on all four graphs: the largest independent HT lies below the smallest clustered HT in each case. Independent sets are also much faster than a single target. On the $n = 16$ barbell the median independent HT is 16.8, against 2558 for one target. Clustering does not reliably reduce the hitting time. On the $n = 16$ barbell the cluster median (2768) lies above the single-target value, with a range of 0.74–1.09 times it; on the 8×8 grid the range is 0.43–1.37 times the single-target value (median 0.89). Whether a cluster is easier or harder than one target depends on where it sits relative to the graph's bottleneck, not only on the number of marked vertices.

Figure 4 places classical hitting times against extended hitting times. For clustered sets and for most independent sets the two are within a few percent. The exception is the independent barbell configuration: 17 of its 18 sets have $\mathrm{HT}^{+}/\mathrm{HT}$ between 1.35 and 1.53, and one set, $\{0, 21, 25, 29\}$, reaches 2.94 (with HT = 415, still six times below the single-target value). So a moderate HT^{+} penalty can arise from a specific independent arrangement without the adversarial construction. The right panel shows the AGJK adversarial torus, where the ratio grows from 5.5 at $L = 18$ to 8.8 at $L = 36$. This is the only family in our data where the penalty grows with size. The growth is consistent with AGJK's $\Omega(N^2)$ statement, but a range of $L \leq 36$ cannot establish the asymptotic rate. For $L = 18$ the interpolated walk still reaches $p_{\max} = 0.978$ at $t = 6$, so the numerical protocol is not slowed down by the large HT^{+} value.

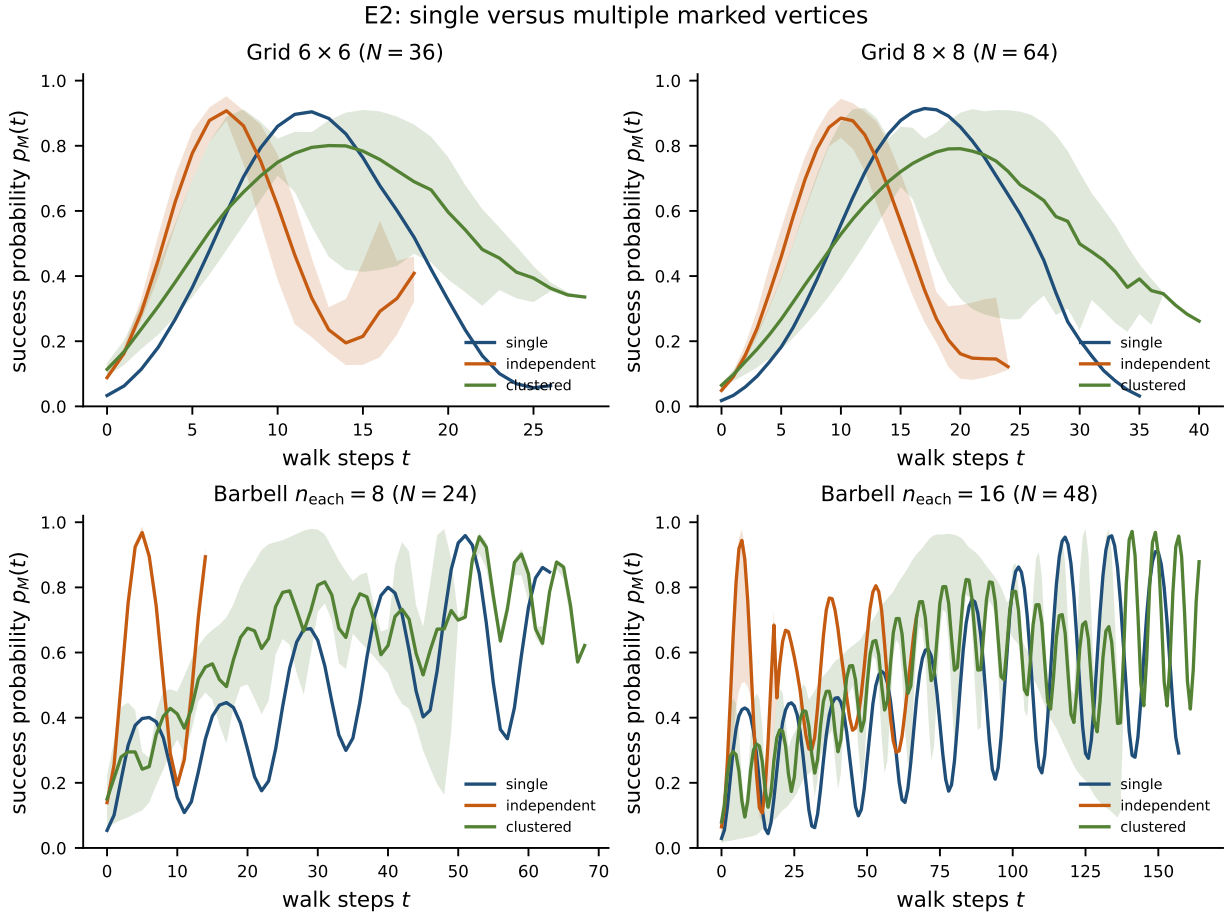


Figure 3: Success probability versus walk steps for single, independent, and clustered marked sets (medians over distinct sets; shaded bands show the min–max range over sets). Independent targets peak earlier and remain easier; clustered targets are slower than independent ones and can be slower or faster than the single target.

Graph	Marked set	sets	median HT [min, max]	median HT^+/HT	max HT^+/HT	median T_q
Grid 8×8	single	1	95.5	1.00	1.00	17
Grid 8×8	independent	18	34.8 [26.9, 39.7]	1.02	1.05	10
Grid 8×8	clustered	18	85.0 [41.5, 131.1]	1.06	1.09	20
Barbell $n = 16$	single	1	2558.0	1.00	1.00	134
Barbell $n = 16$	independent	18	16.8 [16.4, 415.1]	1.53	2.94	7
Barbell $n = 16$	clustered	16	2768.0 [1893.1, 2781.3]	1.00	1.04	133

Table 5: Selected E2 results. Single targets have one set; the other rows summarize all distinct sets from seeds $0, \ldots, 19$.

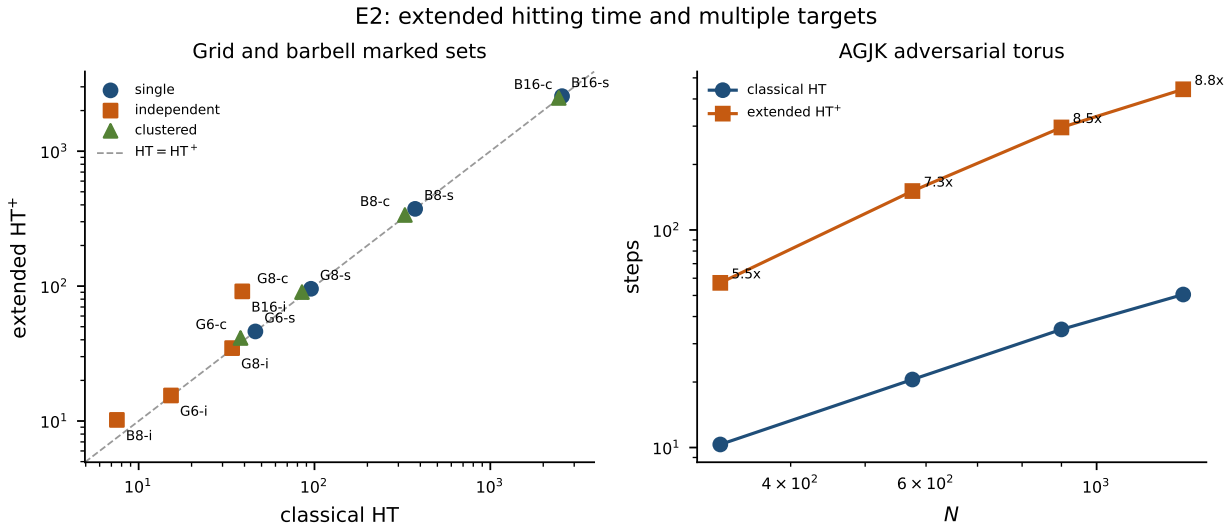


Figure 4: Left: classical hitting time versus extended hitting time for the grid and barbell marked sets; the dashed line is $\mathrm{HT} = \mathrm{HT}^{+}$. Right: the AGJK adversarial torus construction, where HT^{+} grows much faster than HT. Labels give the ratio $\mathrm{HT}^{+}/\mathrm{HT}$.

6.3 Semantic activation (E3)

Figure 5 shows the toy semantic graph and the target activation from the cue `animal` to two targets: `dog` (distance 1) and `violin` (distance 4). The spectral gap is $\lambda_1 = 0.0333$, giving a relaxation time of 30.0 steps. Peaks are taken over $\tau \leq 50$. For the near target the CRW peaks early, at 0.158 with $\tau = 0.13$, and then decays to its stationary value 0.061. The CTQW peak is larger and later, 0.337 at $\tau = 47.5$; the window does not change this peak for $\tau \geq 50$. For the far target the CTQW peaks at 0.268 with $\tau = 19.3$. The CRW has no interior peak: it rises monotonically to its stationary value $\pi_{\text{violin}} = 0.046$, which is reached by $\tau \approx 8$.

The CTQW curves oscillate strongly (fig. 5), so a CTQW peak is the largest of many recurrences and depends on the window; the dog peak at $\tau = 47.5$ is a single spike. The AUC over $\tau \leq 8$ is the more stable comparison.

The activation curves show a clear semantic-distance effect. AUCs are computed over $\tau \leq 8$ so that they do not depend on the longer peak window. For the near target the CRW has a higher AUC (0.579 versus 0.481) because it activates the neighbor immediately and then relaxes. For the far target the CTQW has a higher AUC (0.360 versus 0.323) and a much higher peak, while the CRW never peaks above its stationary value.

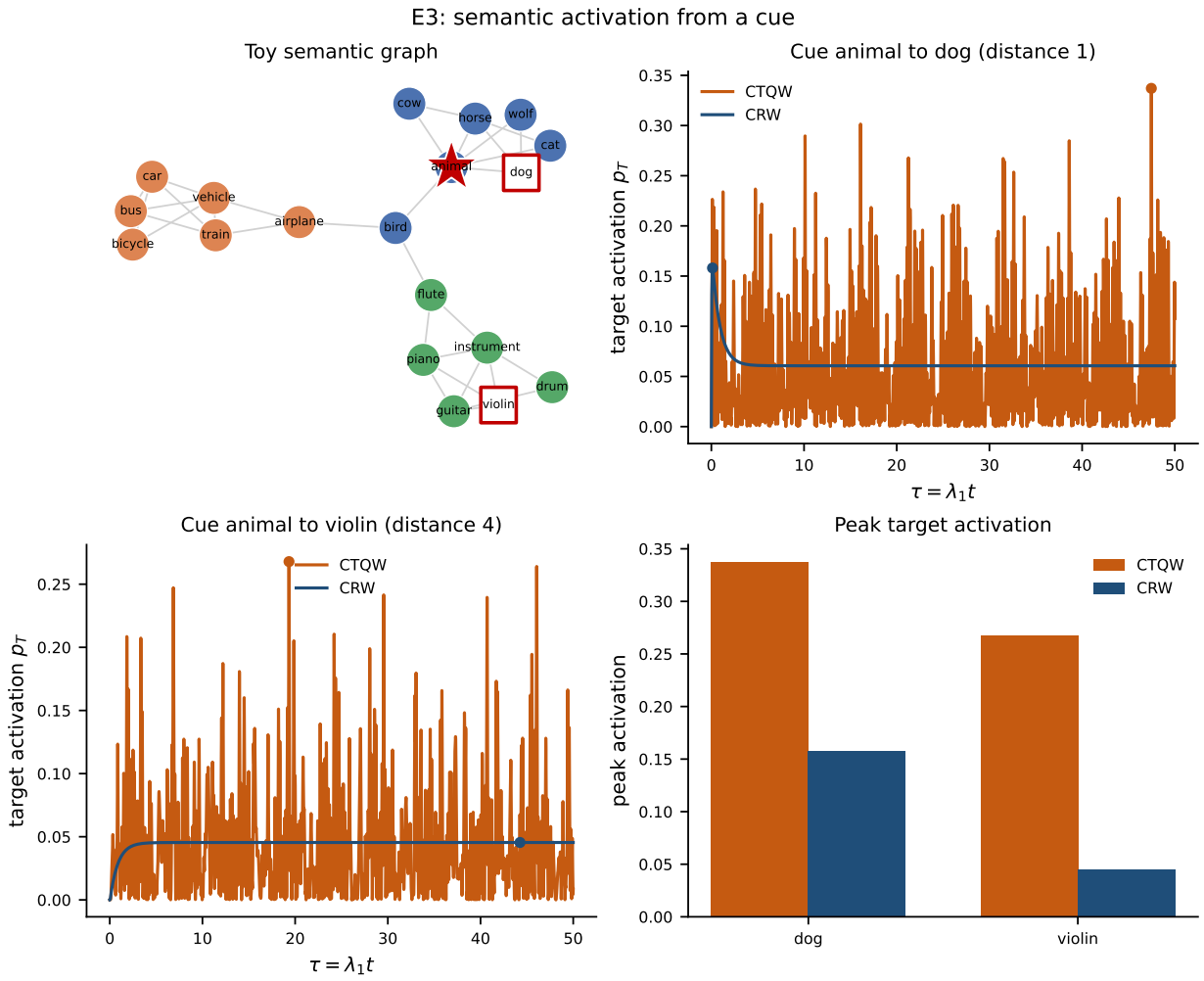


Figure 5: Toy semantic graph and target activation from the cue `animal`. Time is normalized by the spectral gap. The CTQW produces larger and later peaks than the CRW, especially for the cross-community target `violin`.

6.4 Multi-alternative choice under coupling (E4)

The decision experiment has the same pitfall as E5. Our first readout took the choice distribution at the first time the total option probability crossed 0.5. Because the coherent walk oscillates, that time jumps between very different phases when the graph changes slightly. Across the laziness sweep $\eta \in \{0, 0.05, 0.1, 0.2\}$ and ten draws of $\pm 20\%$ edge-weight jitter, the probability of the uncoupled option under this readout moves by a median of 0.44 and up to 0.78 (fig. 6, bottom right), so it cannot support any claim about the effect of coupling. We replace it with two phase-free quantum readouts: *random stopping*, the choice distribution averaged over a uniform stopping time $t \in \{1, \ldots, 400\}$, and *detection*, in which a position measurement at rate $p = 0.2$ per step ends the decision as soon as it lands on an option terminal.

Under the detection readout the result is stable: the spread is at most 0.045, smaller than that of the classical absorbing walk (0.065). The random-stopping readout is in between (median 0.08, maximum 0.26). Both phase-free quantum readouts show the same effect (fig. 6): coupling two

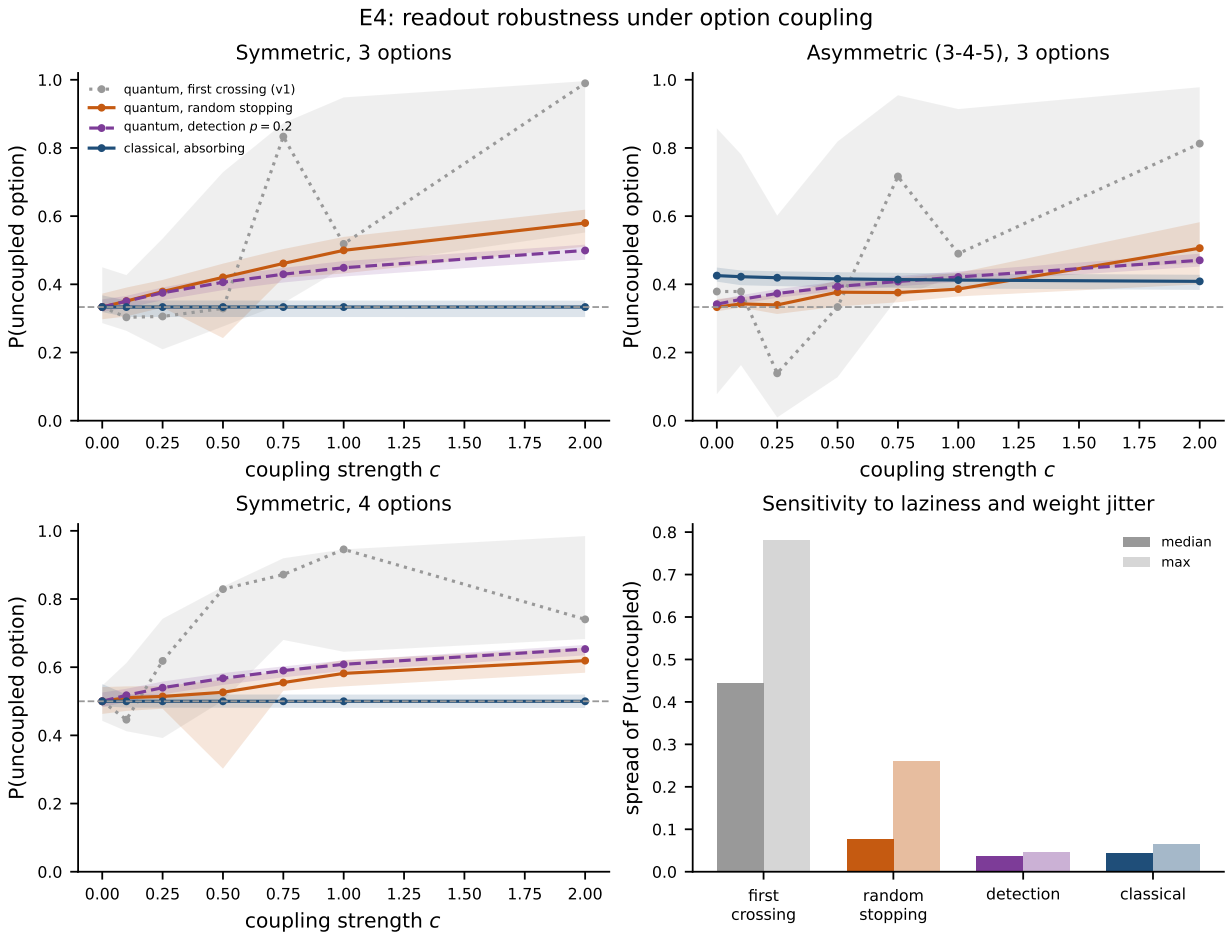


Figure 6: Choice under option coupling. Panels 1–3: probability of the uncoupled option(s) versus coupling c for four readouts; lines are the nominal graph ($\eta = 0.1$), bands span the laziness sweep and edge-weight jitter. The dashed grey line is the uniform baseline. Bottom right: spread of that probability across variants for each readout.

options moves probability to the option that is not coupled, and the shift grows with c. Under detection it is monotone in all three graphs; under random stopping it is monotone up to dips below 0.004 on the asymmetric graph. In the symmetric three-option graph the uncoupled option goes from 1/3 to 0.499 at $c = 2$ under detection (0.580 under random stopping), while the coupled pair falls to 0.250 each. In the four-option graph the two uncoupled options together go from 0.500 to 0.653. In the asymmetric (3–4–5) graph the uncoupled short branch goes from 0.342 to 0.470, monotonically.

The classical walk behaves differently. On the symmetric graphs its choice distribution is exactly uniform for every c: coupling only moves probability between the two coupled branches, which stay symmetric with each other. On the asymmetric graph coupling mostly redistributes mass inside the coupled pair (option B rises from 0.32 to 0.51, option C falls from 0.26 to 0.09) while the uncoupled option changes little (0.426 to 0.409). The quantum effect is therefore a qualitative difference, not a matter of degree. Its cost is time: the mean decision time under detection is 31–44 steps, against 17–27 for the classical walk, because only one step in five is measured.

This is a model-level analogue of the similarity effect in multi-alternative choice: when two

alternatives become more similar, the dissimilar alternative gains share. In the walk model it arises from interference between the coupled branches; the classical walk on the same graph does not produce it.

6.5 Coherence versus intermediate measurement (E5)

A pure quantum walk does not approach its target monotonically; its success probability oscillates. Any comparison at a single deadline therefore depends on where that deadline falls in the oscillation. The top-left panel of fig. 7 makes this explicit for the single-target grid: at $p = 0$ the readout success peaks at 0.449 at $T = 13$ and is only 0.013 at $T = 20$. An earlier version of this analysis compared measurement rates at $T = 20$ only and reported a 12× gain from measurement; that gain came from the deadline sitting in a trough. Under the unified lazy walk ($\eta = 0.1$), which shifts the oscillation, the same comparison gives only 1.6×, and averaging over deadlines removes the gain entirely (below). We therefore report two deadline-free readouts.

Readout model. Here the observer reads the position once at the deadline T, and intermediate measurements only reset the walk. Averaging the success over a uniform deadline $T \in \{1, \ldots, H\}$ removes the phase dependence. With $H = 120$ the pure walk is best or tied in three of four cases (grid single 0.042, grid cluster 0.156, toy single 0.038); only the clustered semantic target gains, from 0.178 to 0.200 at $p = 0.2$. Measurement mainly flattens the curve: the coefficient of variation of p_T drops from 1.50 to 0.37 on the single-target grid at $p = 0.2$. In this model, measurement buys predictability, not success.

Detection model. Here a measurement that lands on a marked vertex ends the search, as in a recognition event, and a final readout is made at the deadline. Now $p = 0$ is the pure walk read once at T and $p = 1$ is exactly the classical first-passage probability $\Pr[\tau_M \leq T]$. Table 6 and the right half of fig. 7 show the result. Before $T \approx 15$ the pure walk wins in every case, because its first coherent peak arrives before measurement has had time to help. For clustered targets an intermediate rate is best for $T \gtrsim 20$, apart from single-step revisits of a later pure-walk peak (e.g. $T = 35$ on the grid): $p^* = 0.2$–0.3 finds the cluster by $T = 40$ with probability 0.455 (grid) and 0.451 (semantic graph), versus 0.326 and 0.342 for the classical walk and 0.055 and 0.158 for the pure walk. By $T = 120$ the advantage over the classical walk is smaller but still present (0.862 versus 0.811; 0.844 versus 0.703). For single targets the optimum moves toward the classical limit as the deadline grows, though not monotonically: on the grid p^* goes from 0 (near the first quantum peak) through 0.2–0.5, with a brief return to 0 at $T = 34$–36, to 1 from $T = 71$; on the semantic graph the classical walk is best for $13 \leq T \leq 85$ and within 0.01 of the best ($p = 0.5$) afterwards.

The picture is a three-way tradeoff. A perfectly timed coherent search has the highest peak, the classical walk accumulates detections reliably, and a partially coherent search combines fast spreading with the ability to stop once the target is seen. The last option wins when the target is extended (a cluster of related concepts) and the deadline is moderate.

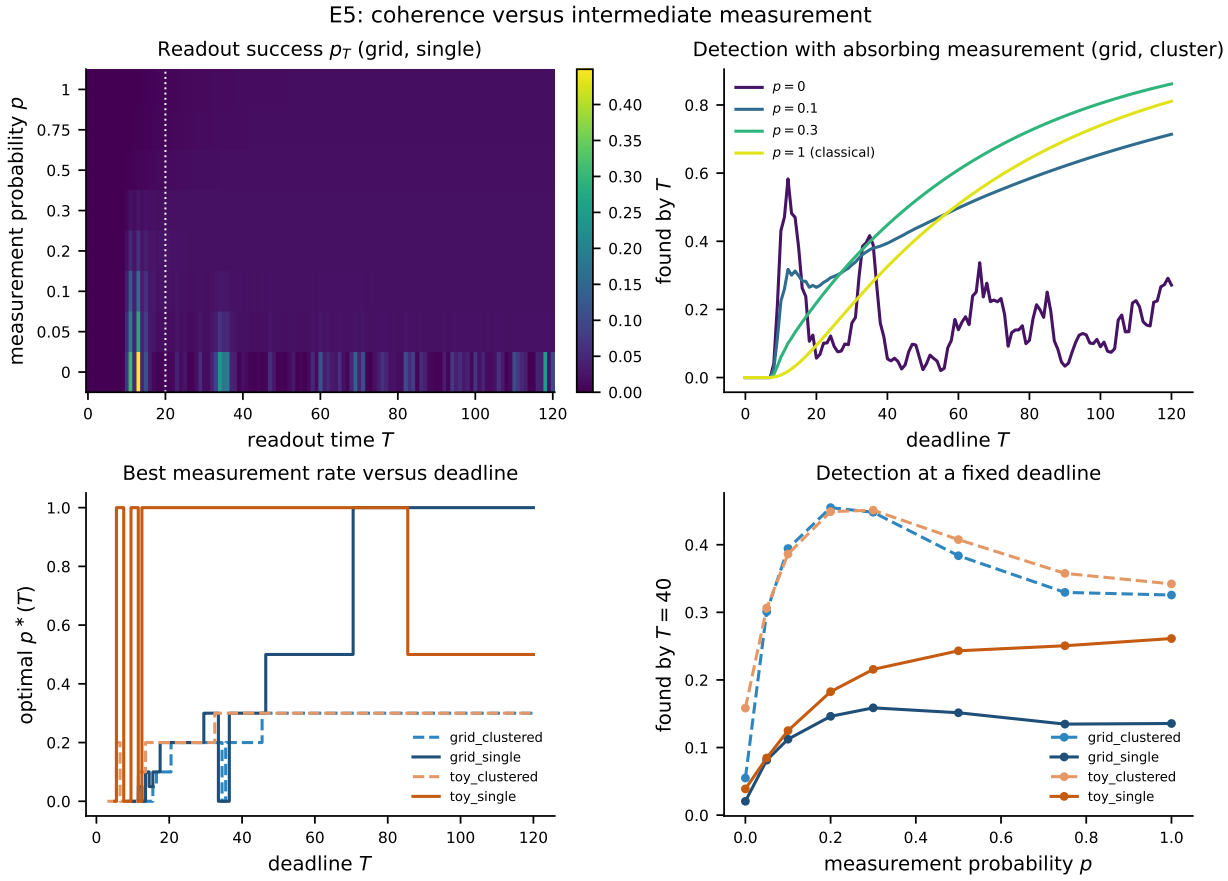


Figure 7: Coherence versus intermediate measurement. Top left: readout success p_T on the single-target grid (rows are the tested p values, not to scale); the dotted line marks the old deadline $T = 20$. Top right: probability that the target cluster has been found by T in the detection model. Bottom left: best measurement rate p^* as a function of the deadline. Bottom right: detection probability at $T = 40$ versus p.

	pure walk ($p = 0$)		found by $T = 40$ (detection model)		
Case	peak	at t	$p = 0$	$p = 1$	best p (value)
Grid, single	0.449	13	0.020	0.136	0.3 (0.159)
Grid, clustered	0.583	12	0.055	0.326	0.2 (0.455)
Toy, single	0.198	107	0.039	0.261	1.0 (0.261)
Toy, clustered	0.394	58	0.158	0.342	0.3 (0.451)

Table 6: E5 summary. The pure walk has the highest peak in every case but needs that peak to be timed; with absorbing measurement an intermediate rate beats both limits for clustered targets.

6.6 Small-world and scale-free graphs (E6)

E1 uses regular graphs. E6 asks whether the $\sqrt{\mathrm{HT}}$ law survives on graphs that resemble semantic networks: Watts–Strogatz small-world graphs with $N \in \{32, 64\}$, $k = 4$, and rewiring probability $\beta \in \{0, 0.05, 0.2, 1\}$, and Barabási–Albert scale-free graphs with $m = 2$. Each configuration uses five random seeds. The marked vertex is either the highest-degree ("hub") or the lowest-degree ("peripheral") vertex, giving 100 graph–target instances in total. Over all instances the quantum search time follows $T_q \sim \mathrm{HT}^{0.52}$ (95% CI $[0.49, 0.54]$, $R^2 = 0.94$). The exponent is slightly above $1/2$ on the small-world family (0.54, 80 instances) and slightly below on the scale-free family (0.45, 20 instances).

Rewiring shortens classical hitting times for hub targets from 91 to 25 ($N = 32$) and from 334 to 52 ($N = 64$). Their quantum speedup falls from 4.3 to 2.7 and from 8.0 to 4.0. Peripheral targets are harder classically, and their speedup stays between about 4 and 9, with a non-monotone dependence on β. On scale-free graphs the contrast is largest: hub targets have HT $\approx$ 11–20 and speedup 1.8–2.4, while peripheral targets have HT $\approx$ 143–298 and speedup 6.8–11.5. Across all instances $\log T_q$ tracks $\log \mathrm{HT}$ closely (correlation of $\log \mathrm{HT}$ and $\log(\mathrm{HT}/T_q)$ is 0.96). The quantum advantage is therefore largest where the classical search is hardest, which is the same pattern as E1.

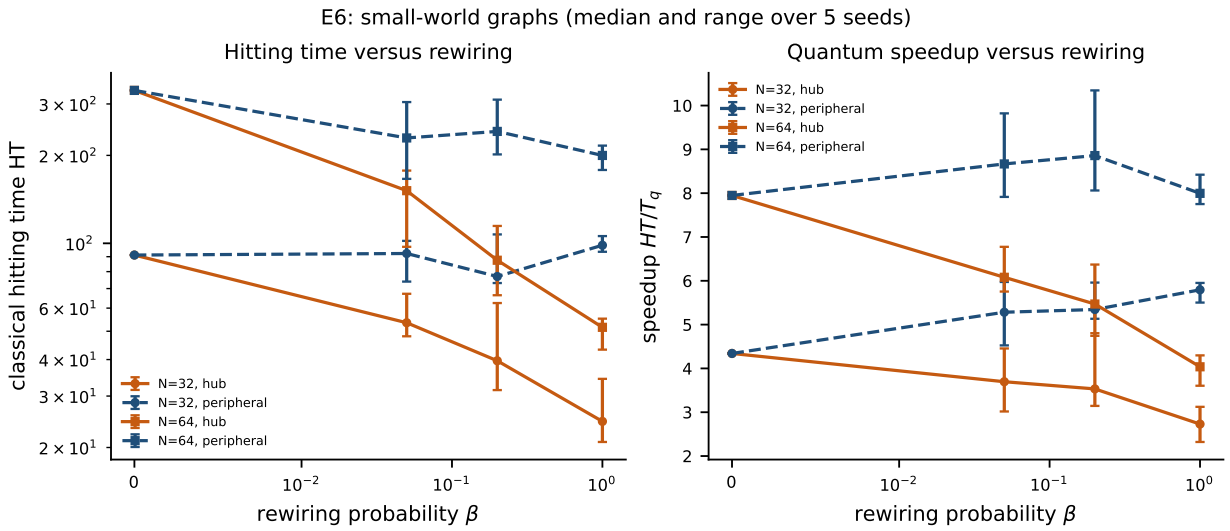


Figure 8: Small-world graphs. Left: classical hitting time versus rewiring probability. Right: speedup HT/T_q. Points are medians and bars give the range over five seeds. At $\beta = 0$ the ring is regular, so hub and peripheral targets coincide.

6.7 Summary of empirical findings

The experiments support three conclusions. First, the quantum search time tracks $\sqrt{\mathrm{HT}}$ across the synthetic families, and independent targets are easier than clustered ones. Second, a large $\mathrm{HT}^+/\mathrm{HT}$ gap is a structural effect of the marked-set arrangement, not a generic consequence of having multiple targets. Third, the cognitive layer is sensitive to graph structure: semantic distance changes activation timing, option coupling shifts quantum choice toward the uncoupled option while leaving the classical choice unchanged, and when a measurement can end the search, an intermediate measurement rate finds clustered targets faster than either the pure quantum or the classical walk, even though the pure walk keeps the highest timed peak.

7 Discussion

7.1 What the experiments show

The scaling experiments reproduce the central AGJK phenomenon. Across four graph families the quantum search time fits $T_q \sim \mathrm{HT}^{0.44\text{–}0.52}$, close to the predicted square-root law, and the peak success probability is high enough that the fitted times are meaningful. The cycle gives the largest speedup because its classical hitting time is almost quadratic. The complete graph and hypercube have smaller finite-size speedups because their classical hitting times grow almost linearly over the simulated range.

The multiple-target experiments refine the picture. Independent targets are easier: every independent set we tried was easier than every clustered set, and spreading targets across the graph lowers both the classical hitting time and the quantum search time. Clustered targets are not uniformly easier or harder than a single target. Their hitting time depends on whether the cluster sits near the graph's bottleneck, because reaching the cluster does not mean finding a particular vertex. The large $\mathrm{HT}^+ \gg \mathrm{HT}$ gap appears in the periodic adversarial construction of AGJK, where it grows with size. A moderate gap of 2.9 also appears for one specific independent set on a barbell

graph, so multiple targets alone do not create a large penalty, but a specific arrangement can produce a moderate one. The penalty is a property of the arrangement and of the resulting spectral structure.

The semantic experiment shows a distance-dependent difference between CTQW and CRW. For a near target the classical walk activates the neighbor immediately and has a larger early AUC. For a far target the quantum walk maintains a larger and later peak, and its activation is spread over a larger radius. This is consistent with the intuition that coherent propagation can bridge communities, while classical diffusion relaxes to the stationary distribution. The result is a model-level prediction: a semantic-search experiment should see a larger peak but a later response for cross-community targets under the quantum model, and an earlier but lower activation under the classical model.

The decision experiment produces a context effect that the classical walk cannot. Coupling two options shifts quantum choice toward the uncoupled option, increasingly with the coupling strength, while the classical choice on symmetric graphs stays exactly uniform. The effect is moderate (from $1/3$ to about 0.5 on the symmetric three-option graph). Under the detection readout it is monotone and robust to laziness and edge-weight noise (spread ≤ 0.045); under random stopping it is noisier; under a first-threshold readout it is swamped by the oscillation phase. This is the same lesson as in E5: a coherent walk must be read out with a stopping rule that does not depend on its phase.

The decoherence experiment answers the noise question only once the readout is specified. Comparing rates at one deadline is unreliable, because the pure walk oscillates and the comparison then measures the oscillation phase; our own first analysis made this mistake and reported a $12\times$ gain that disappears under deadline averaging. If measurement only resets the walk, it makes the success curve flatter but not higher. If a measurement can also end the search, the pure walk wins only at short deadlines that catch its first peak; beyond that an intermediate rate $p \approx 0.2$–0.3 beats both the pure quantum and the classical walk for clustered targets, and for single targets the best rate moves, non-monotonically, toward the classical limit. The model therefore predicts that the useful amount of "introspection" depends on how extended the target is and how much time is available, which is a testable prediction for deadline-manipulation experiments.

7.2 Relation to the literature

The technical results are consistent with AGJK (Ambainis et al., 2019). Our numerical protocol tests the simpler interpolated-walk algorithm (Conjecture 3) and cites the fast-forwarding theorem as the proved guarantee. The adversarial torus reproduces the qualitative $\mathrm{HT}^{+} \gg \mathrm{HT}$ behavior and shows that the effect survives at modest sizes ($L = 18$–36). We do not claim a new theorem; the contribution is the numerical validation and the cognitive interpretation.

The semantic experiment is closest to Pavone and Faro (Pavone and Faro, 2025), but it uses a discrete marked-vertex search perspective rather than an insight task. The decision experiment is related to Huang and Busemeyer (Huang and Busemeyer, 2022) and to the evidence-accumulation literature (Chen et al., 2022). The decoherence model is a simplified version of the open-system ideas in Epping et al. (Epping et al., 2023): projective measurements at a fixed rate rather than a Lindblad master equation. The qualitative conclusion—that noise can help at intermediate levels and hurt at strong levels—is consistent with the broader quantum-cognition literature on decoherence-assisted processes.

7.3 Algorithmic versus biological claims

The project is deliberately agnostic about biology. Quantum walks are used as a mathematical process model, not as a claim that the brain is a quantum computer. The algorithmic advantage is a statement about the number of walk steps needed to reach a target under a particular graph and marked set. Whether a biological system can realize such a walk is a separate question. We keep the two claims separate throughout: the QIS results concern complexity and spectra, and the cognitive results concern the qualitative patterns that a process model can produce.

This separation also limits the interpretation of the context effects. The decision experiment shows that a quantum walk can produce a similarity-like shift in choice under option coupling. It does not show that human similarity effects arise from quantum interference. The value of the model is that it generates distinct predictions: a classical random walk and a quantum walk on the same graph produce different choice distributions and different response times, so an experiment could in principle distinguish them.

7.4 Limitations

Several limitations should be stated clearly.

- **No human data.** The experiments are numerical. The cognitive quantities are model-defined, not fitted to behavioral data.
- **Synthetic graphs.** The semantic graph is hand-built and small (19 vertices). It illustrates the mechanism but does not capture the scale or noise of real semantic memory.
- **Interpolation selection.** The parameter s is chosen by a grid search on the AGJK lower bound. A different selection rule could change the constants in the search times, although the scaling exponents are stable.
- **Two walk formalisms.** The search experiments use discrete-time Szegedy walks, while the semantic experiment uses continuous-time walks. We compare them only within their own time normalizations and do not mix the two formalisms in a single fit.
- **Simplified noise.** The decoherence model uses projective position measurements and resets. It is not a full Lindblad or open-system model.
- **Finite sizes.** The graphs have at most a few hundred vertices. The asymptotic exponents are estimated from small ranges, especially for the grid and hypercube.

7.5 Future work

Four extensions would strengthen the project. First, replace the toy semantic graph with a ConceptNet subgraph and test whether the distance-dependent activation pattern survives in a real semantic network. Second, fit the decision model to behavioral choice and response-time data, using the coupling parameter as a free similarity parameter. Third, replace the projective measurement model with a Lindblad equation or a continuous dephasing model and compare the optimal noise level. Fourth, implement the decision graph on a small quantum device or circuit simulator to test whether the context effect survives realistic noise. A larger graph study could also test how the size of the context effect depends on branch lengths, coupling patterns, and the measurement rate of the detection readout.

8 Conclusion

This project used the AGJK marked-vertex search algorithm as the QIS core of a process-level model of cognitive search. We implemented the Szegedy and interpolated walks, verified the $\widetilde{O}(\sqrt{\mathrm{HT}})$ scaling on four graph families, including small-world and scale-free graphs, and compared single, independent, and clustered targets. The main structural finding is that every independent set is easier than every clustered set, while clustered sets can be easier or harder than a single target. The large $\mathrm{HT}^{+} \gg \mathrm{HT}$ gap requires the adversarial periodic arrangement of AGJK; a moderate gap also appears for one independent set on a barbell graph. On the cognitive side, a toy semantic graph produced distance-dependent activation patterns, a multi-branch decision graph produced a robust similarity-like context effect under option coupling that the classical walk lacks, and a partial-measurement model showed that, beyond the shortest deadlines, an intermediate measurement rate that can stop the search finds clustered targets faster than both the pure quantum and the classical walk.

The broader lesson is that the algorithmic structure of quantum walk search is a useful source of cognitive hypotheses. The marked-set structure, the extended hitting time, and the success curve are not merely complexity quantities; they define a process model with testable predictions. At the same time, the project is deliberately cautious: the results are numerical, model-level, and do not claim a biological quantum advantage. The next step is to test the same predictions on larger semantic networks and, eventually, on behavioral choice and response-time data.

AI Tool Usage

An AI coding assistant was used for code scaffolding, experiment automation, and LaTeX drafting. All technical derivations, experiment design, numerical results, and interpretations were checked by the authors. The AI tool was not used to generate research results or citations without verification.

A Hitting-time derivations

Complete graph. For K_N with one marked vertex, every unmarked vertex is equivalent. Let h be the expected hitting time from an unmarked vertex. Then

$$h = 1 + \frac{N-2}{N-1}h, \tag{31}$$

so $h = N - 1$. Since the stationary distribution is uniform and we condition on starting outside the marked set, $\mathrm{HT} = N - 1$.

Cycle. For C_N with marked vertex 0, the expected hitting time from distance i is $h_i = i(N-i)$. It satisfies the recurrence

$$h_i = 1 + \frac{1}{2}h_{i-1} + \frac{1}{2}h_{i+1}, \qquad h_0 = h_N = 0, \tag{32}$$

whose solution is the quadratic above. Averaging over $i = 1, \ldots, N-1$ gives

$$\mathrm{HT} = \frac{1}{N-1}\sum_{i=1}^{N-1} i(N-i) = \frac{N(N+1)}{6}. \tag{33}$$

These identities are used as sanity checks in `tests/test_classical.py`.

B Discriminant matrix and spectral identity

For a reversible chain, detailed balance gives $\pi_x P_{xy} = \pi_y P_{yx}$, hence

$$D_{xy} = \sqrt{P_{xy}P_{yx}} = \frac{\sqrt{\pi_x}P_{xy}}{\sqrt{\pi_y}}, \tag{34}$$

so $D = \mathrm{diag}(\pi)^{1/2} P\, \mathrm{diag}(\pi)^{-1/2}$. The matrices D and P are similar, so they have the same eigenvalues, and D is symmetric. The identity

$$\langle\bar{0}|\langle x|W(P)|\bar{0}\rangle|y\rangle = \sqrt{P_{xy}P_{yx}} = D_{xy} \tag{35}$$

follows directly from the definitions of $V(P)$, Shift, and Ref. For the interpolated walk, the same computation gives the top-left block $D(s)$, which is the starting point for the Chebyshev bound (13).

C Measurement ensemble recursion

Let $|\phi_{x,a}\rangle = W(0)^a|\bar{0}\rangle|x\rangle$ for $a = 0, \ldots, T_{\max}$, with position marginal $\mu_{x,a}(y) = \|\Pi_y|\phi_{x,a}\rangle\|^2$ and marked mass $s_{x,a} = \sum_{y\in M}\mu_{x,a}(y)$. Let $\mu_a^{(0)}$ and $s_a^{(0)}$ be the corresponding quantities for the initial state. A trajectory is described by $q_0(t)$, the probability that no measurement has occurred by time t, and by $q_{x,a}(t)$, the probability that the last reset was at vertex x and the state has evolved for a steps since that reset.

At one step,

$$q_0(t+1) = (1-p)q_0(t), \tag{36}$$

$$q_{x,a+1}(t+1) = (1-p)q_{x,a}(t), \tag{37}$$

$$q_{y,1}(t+1) = p\, q_0(t)\mu_t^{(0)}(y) + p\sum_{x,a} q_{x,a}(t)\mu_{x,a}(y). \tag{38}$$

The success probability is

$$p_M(t+1) = q_0(t+1)s_{t+1}^{(0)} + \sum_{x,a} q_{x,a}(t+1)s_{x,a}. \tag{39}$$

This recursion is exact for the channel (16). It requires the basis ensembles once per graph and marked set, after which a sweep over p is inexpensive. The implementation is in `src/noise.py`.

For the detection model, the mass that a measurement places on M is removed instead of reset. With $f(t)$ the probability found by time t,

$$f(t+1) = f(t) + p\sum_{y\in M}\Big[q_0(t)\mu_t^{(0)}(y) + \sum_{x,a} q_{x,a}(t)\mu_{x,a}(y)\Big], \tag{40}$$

the reset term $q_{y,1}(t+1)$ is restricted to $y \notin M$, and the probability of having found the target with a final readout at T is $F_p(T) = f(T) + q_0(T)s_T^{(0)} + \sum_{x,a} q_{x,a}(T)s_{x,a}$.

D Additional results

Figure 9 shows the classical-to-quantum speedup for E1. The cycle has the largest speedup because its classical hitting time grows almost quadratically; the complete graph has the smallest because its classical hitting time grows only linearly.

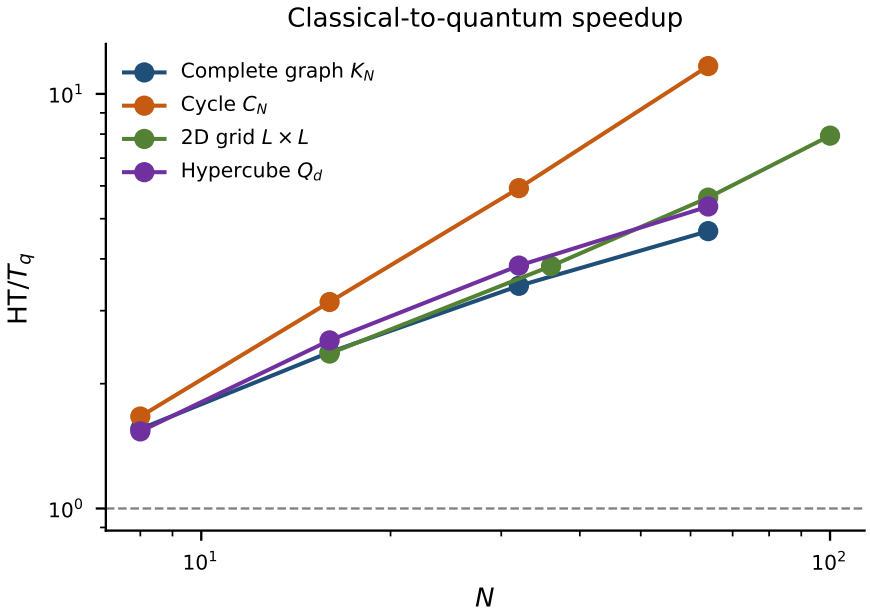


Figure 9: Classical-to-quantum speedup HT/T_q versus graph size for E1.

E Reproducibility

The environment is created with

```
mamba env create -f environment.yml
mamba activate qwalk-cognition
```

from `Final/code`. Everything is reproduced from `Final/code` with

```
make all
```

which runs the tests, all six experiments, copies the figures used in the paper, writes `results/MANIFEST.json` (git commit, dirty-tree flag, package versions, and SHA-256 checksums of configs and results), and compiles the report. Single experiments can be run with, for example, `python -m src.experiments -only E6`. The complete run takes about fifteen minutes, most of it in E4's robustness sweep. The test suite contains 71 passing tests. The configuration files are `configs/base.yaml` and `configs/experiments.yaml`; all random choices use explicit seeds.

References

Andris Ambainis, András Gilyén, Stacey Jeffery, and Martins Kokainis. Quadratic speedup for finding marked vertices by quantum walks. *arXiv preprint arXiv:1903.07493*, 2019.

Jerome R. Busemeyer and Peter D. Bruza. *Quantum Models of Cognition and Decision*. Cambridge University Press, 2 edition, 2024. doi: 10.1017/9781009205351.

Ming Chen, Giuseppe M. Ferro, and Didier Sornette. On the use of discrete-time quantum walks in decision theory. *PLoS ONE*, 17(8):e0273551, 2022. doi: 10.1371/journal.pone.0273551.

Graph	Marked set	sets	median HT [min, max]	median HT^+/HT	max HT^+/HT	median T_q
Grid 6×6	single	1	46.1	1.00	1.00	12
Grid 6×6	independent	14	15.7 [12.7, 17.4]	1.01	1.02	7
Grid 6×6	clustered	17	40.1 [17.6, 54.9]	1.09	1.10	14
Grid 8×8	single	1	95.5	1.00	1.00	17
Grid 8×8	independent	18	34.8 [26.9, 39.7]	1.02	1.05	10
Grid 8×8	clustered	18	85.0 [41.5, 131.1]	1.06	1.09	20
Barbell $n = 8$	single	1	373.6	1.00	1.00	51
Barbell $n = 8$	independent	11	7.6 [7.1, 7.6]	1.38	1.38	5
Barbell $n = 8$	clustered	9	425.3 [196.6, 433.2]	1.01	1.09	48
Barbell $n = 16$	single	1	2558.0	1.00	1.00	134
Barbell $n = 16$	independent	18	16.8 [16.4, 415.1]	1.53	2.94	7
Barbell $n = 16$	clustered	16	2768.0 [1893.1, 2781.3]	1.00	1.04	133

Table 7: Full E2 results. Single targets have one set; other rows summarize all distinct sets from seeds $0, \ldots, 19$.

Gijs P. Epping, Peter D. Kvam, Timothy J. Pleskac, and Jerome R. Busemeyer. Open system model of choice and response time. *Journal of Choice Modelling*, 49:100453, 2023. doi: 10.1016/j.jocm.2023.100453.

Jiaqi Huang and Jerome R. Busemeyer. A quantum walk framework for multialternative decision making. *Proceedings of the 44th Annual Meeting of the Cognitive Science Society*, 2022. doi: 10.31234/osf.io/hmv4a.

Arianna Pavone and Simone Faro. Modeling the emergence of insight via quantum interference on semantic graphs. *Mathematics*, 13(19):3171, 2025. doi: 10.3390/math13193171.

Emmanuel M. Pothos and Jerome R. Busemeyer. Quantum cognition. *Annual Review of Psychology*, 73:749–778, 2022. doi: 10.1146/annurev-psych-033020-123501.

Dominic Widdows, Jyoti Rani, and Emmanuel M. Pothos. Quantum circuit components for cognitive decision-making. *Entropy*, 25(4):548, 2023. doi: 10.3390/e25040548.